A Quantity Theory of Money Approach to Cryptocurrency Valuation: Store of Value, Medium of Exchange, and the Economic Moat Signal

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June 2026

Preliminary. Comments welcome.

Generated with AI assistance under AFA 2027 Special Session guidelines.

Abstract

We apply the Quantity Theory of Money (QTM) to the cross-sectional valuation of cryptocurrencies and protocol tokens. Treating each digital asset as the currency of a self-contained economy, we decompose market capitalization into a Medium of Exchange (MoE) component, priced by the free-circulating velocity of supply, and a Store of Value (SoV) residual. The key innovation is λ—the fraction of total supply that is locked or staked—which partitions the circulating supply into economically active and inert components. A central result is that the SoV/MoE ratio simplifies to $\lambda/(1-\lambda)$, directly observable from on-chain data without velocity estimation. For native blockchain coins, high λ reflects a convenience yield and seigniorage signal grounded in monetary

search theory. For governance and utility tokens, high λ is an economic moat signal, capturing token holder fundamental conviction in platform growth. We show that $\lambda/(1-\lambda)$ alone positively predicts future cross-sectional returns, and that this predictability is substantially stronger when the asset is cheap relative to its fundamental economic activity—measured by our Growth-Levelized NVT ratio—supporting the interpretation that NVT identifies when the conviction signal is not yet priced in. A long Stars/short Avoid portfolio generates positive factor-adjusted alpha after controlling for market, size, and momentum factors.

1 Introduction

What determines the value of a cryptocurrency? The dominant approach in the literature has been to adapt discounted cash flow (DCF) models, searching for a dividend-like flow—transaction fees, staking rewards, governance rights—that can be capitalized into a present value (Liu and Tsyvinski, 2021; Pagnotta and Buraschi, 2022; Biais, Bisière, Bouvard, and Casamatta, 2019). This approach, however, mischaracterizes the fundamental nature of digital assets. Cryptocurrencies and protocol tokens exhibit currency-like behavior: each coin or token is the monetary instrument of a self-contained digital economy, and its value is governed by the same forces that determine the value of any currency—the scale of economic activity it enables and the efficiency with which it circulates. The correct framework, we argue, is the Quantity Theory of Money (Fisher, 1911).

The Fisher equation, $MV = PQ$, relates the stock of money (M) and its velocity (V) to the nominal value of economic output (PQ) (Fisher, 1911). Applied to digital assets, a cryptocurrency's market capitalization proxies its money supply, and the economic output it supports—on-chain transaction volume for native coins, protocol throughput for tokens—is its PQ. This is not merely an analogy. Cong, Li, and Wang (2021) develop a formal dynamic model in which token prices aggregate heterogeneous users' transactional demand, with equilibrium prices governed by platform adoption—a formulation directly consistent with the QTM framework we develop here. Sockin and Xiong (2020) similarly model tokens as platform membership instruments, showing that the token price reflects the present value of the convenience yield accruing to participants: the non-pecuniary benefit of network membership, including access to platform services, network liquidity, and future transactional capability.

Before proceeding, we define the two components of value our framework decomposes. The *Medium of Exchange* (MoE) value of a digital asset is the value attributable to its current transactional function: the economic activity it enables today, priced by the velocity with which it circulates. The *Store of Value* (SoV) value is the premium agents are willing to pay today by forgoing current exchange utility—the transactions, goods, services, or alternative investments they could obtain by spending the asset now—in anticipation of future value appreciation. This appreciation is expected to arise from network growth, expanding user adoption, increasing utility, and innovation. An agent holding Ethereum rather than spending it is implicitly paying an intertemporal price: she sacrifices today's exchange utility for her conviction that the network's future worth exceeds today's use value. The SoV component is the aggregate market valuation of this forward-looking conviction.

The QTM framework requires a critical extension to operationalize this decomposition. Not all of a cryptocurrency's circulating supply is economically active. A substantial and growing fraction is locked in staking contracts, pledged as governance collateral, or held in long-term custody with near-zero velocity. These holdings are economically inert from a transactional standpoint: they contribute to the money supply but perform no current exchange function. Conflating active and inactive supply distorts velocity estimates and, consequently, misprices the economic content of market capitalization. We address this by introducing λ, defined as the proportion of the total supply that is locked or staked—including tokens committed to network validation, governance contracts, and other long-term holding functions. The formal derivations are provided in Appendix A. The key result is that the SoV/MoE ratio simplifies to $\lambda/(1-\lambda)$: a monotone, directly observable function of λ alone, requiring no velocity estimation.

Why does λ matter beyond its role in the decomposition? For native blockchain coins—Ethereum, Solana, and their peers—high λ is grounded in three complementary theoretical traditions. First, *monetary search theory* (Kiyotaki and Wright, 1993; Lagos and Wright, 2005) establishes that money emerges as a coordination equilibrium: an asset becomes valuable because rational agents expect others to hold and accept it. High λ is the blockchain-observable, publicly verifiable signal of this coordination commitment. Second, the *convenience yield* (Sockin and Xiong, 2020; Krishnamurthy and Vissing-Jorgensen, 2012) is the non-pecuniary benefit agents derive from holding a monetary asset: access to network security, platform liquidity, and future transactional capability. Just as holders of U.S. Treasury securities accept lower yields in exchange for liquidity services (Nagel, 2016), holders of dominant blockchain coins accept lower required returns because network membership has value beyond current transactions. Crucially, the SoV premium is the present value of expected *future* convenience yields: agents who lock coins today are paying for anticipated growth in the network's membership value. Third, the *seigniorage* channel (Cong and He, 2024) anchors this signal in rational optimization: in proof-of-stake networks, newly minted tokens distributed to stakers constitute monetary seigniorage, and the staking decision reveals belief that this income plus anticipated price appreciation exceeds the opportunity cost of locking capital. These three channels are complementary: coordination theory explains why high λ is self-reinforcing, convenience yield provides the holding motive, and seigniorage grounds the rational optimization.

Governance and utility tokens present a related but distinct case. A governance token such as UNI is the monetary instrument of the Uniswap protocol economy. Its economic output is the total throughput of the Uniswap protocol—decentralized exchange volume, liquidity deployed, fees generated. Whether protocol revenue currently accrues to token holders is immaterial, just as it is immaterial whether U.S. GDP flows to dollar holders. This framing aligns with Cong, Li, and Wang (2021), who price tokens on the present value of future platform utility and adoption, and Schilling and Uhlig (2019), who model the value of digital currencies as functions of the economic activity they support. For governance tokens, λ carries a different signal. Staking a governance token into a DAO contract renders it illiquid: staked tokens cannot be sold while locked. An agent who voluntarily incurs this illiquidity cost reveals a preference for future appreciation over present liquidity—a choice grounded in assessments of network effects, utility expansion, competitive positioning, and innovation trajectory. Drawing on Hirschman (1970), holders who exercise *voice* (governance participation) rather than *exit* (selling) reveal a preference for the protocol's future that passive buyers do not. Edmans and Manso (2011) formalize this in a corporate setting:

blockholders who engage rather than exit create governance value precisely because engagement is costly and therefore credible. Applied to tokens, active governance participation is a costly signal (Spence, 1973) of fundamental conviction in platform growth, making high participation rates an economic moat indicator.

We operationalize these ideas in two metrics. The *SoV/MoE ratio*—equal to $\lambda/(1-\lambda)$—measures the balance between forward-looking conviction and current transactional utility, and is our primary return predictor. The *Growth-Levelized NVT ratio* normalizes market capitalization by the present value of the network's expected transaction stream (constructed analogously to a PMT-based discounted cash flow, as detailed in Section 3), yielding a measure of whether the asset is cheap or expensive relative to its fundamental economic activity. Crucially, Growth-Levelized NVT does not carry its own independent theoretical prediction for returns. Rather, it serves as a conditioning variable: high $\lambda/(1-\lambda)$ is most valuable as a return predictor when the market has not yet priced in the underlying conviction signal—that is, when Growth-Levelized NVT is low. Assets that combine high conviction with cheap valuation (Stars) offer the best expected returns; assets that combine low conviction with expensive valuation (Avoid) offer the worst. We find that $\lambda/(1-\lambda)$ alone positively predicts future returns cross-sectionally for both coins and governance tokens. The predictive power is substantially stronger for assets with below-median Growth-Levelized NVT, consistent with the view that NVT identifies when the conviction signal remains unpriced. A long Stars/short Avoid calendar-time portfolio generates an annualized alpha of [X]% with a Sharpe ratio of [Y] after controlling for market, size, and momentum factors, with the results holding separately for coins and governance tokens.

Related Literature. A growing body of work attempts to value digital assets, and our paper connects to several strands. On equilibrium cryptocurrency valuation, Pagnotta and Buraschi (2022) model Bitcoin value as a function of user adoption and network security; Biais, Bisière, Bouvard, and Casamatta (2019) study equilibrium selection in blockchain protocols; Schilling and Uhlig (2019) develop a general equilibrium model of Bitcoin as a competing currency; and Easley, O'Hara, and Basu (2019) analyze how transaction fees connect mining incentives to market value. Cong, Li, and Wang (2021) and Sockin and Xiong (2020) are the closest to our approach in modeling token prices through adoption dynamics and convenience yields respectively; we complement them by providing an empirically tractable QTM decomposition and a testable investment signal derived from on-chain observables. On crypto risk factors and return predictability, Liu and Tsyvinski (2021) identify network, momentum, and investor attention factors; Liu, Tsyvinski, and Wu (2022) develop a three-factor model for cryptocurrency returns; and Makarov and Schoar (2020) document price dislocations across exchanges. We contribute to this literature the first on-chain supply-side signal—$\lambda/(1-\lambda)$—grounded in monetary theory rather than return-based factor construction. On the monetary economics of digital assets, Kiyotaki and Wright (1993), Lagos and Wright (2005), and Schilling and Uhlig (2019) provide search-theoretic and general equilibrium foundations; Cong and He (2024) analyzes proof-of-stake tokenomics; and Jiang, Krishnamurthy, and Lustig (2024) draw the analogy between dollar exorbitant privilege and dominant-network premia. The convenience yield literature (Krishnamurthy and Vissing-Jorgensen, 2012; Nagel, 2016) motivates the monetary premium in $\mathrm{MC}_{\mathrm{SoV}}$. On token issuance and governance, Howell, Niessner, and Yermack (2020) study the determinants of ICO success; Chod and Lyandres (2021) model ICO design under information asymmetry; and Edmans and Manso (2011), Hirschman (1970), and DAO Governance (2024) provide the governance theory and evidence linking participation to token value. Our paper is the first, to our knowledge, to derive a directly observable valuation decomposition from QTM, to distinguish the signal content of λ for coins versus governance tokens, and to demonstrate that the interaction of conviction and valuation predicts cross-sectional cryptocurrency returns.

The remainder of the paper is organized as follows. Section 2 develops the theoretical framework and states the paper's testable hypotheses. Section 3 describes the data and variable measurement. Section 4 presents the main results. Section 5 reports robustness tests. Section 6 concludes.

2 Theoretical Framework and Hypothesis Development

The QTM decomposition in Appendix A is an accounting identity: given λ, the SoV/MoE ratio follows mechanically. It is not, by itself, a behavioral prediction. This section asks why λ, and its interaction with valuation, should predict future returns. We develop two stylized one-period models—one for the locking decision common to coins and governance tokens, and one for the price-discovery role of Growth-Levelized NVT—each built directly on the theoretical traditions introduced in Section 1. Each model yields a proposition, stated formally as a hypothesis, that we take to the data in Section 4.

2.1 The Locking Decision

Consider a continuum of agents, indexed by $i \in [0, 1]$, who hold the native asset of a self-contained digital economy at date t—the native coin of a blockchain network, or the governance token of a protocol built on it—and choose whether to lock it for one period (stake, or commit it to a governance contract) or hold it freely circulating. The derivation below is for a single, fixed asset; we suppress an explicit asset index throughout, since the agent-level mechanism below is identical for any asset held fixed, with all asset-level heterogeneity entering only through θ_{t+1} and b_t. The empirical content of the model is fundamentally a cross-sectional claim across assets, which we make explicit when restating it as Proposition 1 below.

Locking pays a current, publicly observed benefit $b_t \geq 0$: for proof-of-stake coins, newly issued seigniorage distributed to validators (Cong and He, 2024); for governance tokens, a direct fee or yield share distributed to stakers where the protocol has one, and $b_t = 0$ for protocols with no revenue-sharing mechanism. Locking also carries a convenience yield realized at $t + 1$, the non-pecuniary benefit of network or protocol membership emphasized by Sockin and Xiong (2020): validation or governance rights, platform liquidity, and anticipated future transactional capability. We decompose agent i's convenience yield into a common component θ_{t+1}—the asset's true future growth trajectory, unobserved at t—and an idiosyncratic, mean-zero noise term ε_i reflecting agent i's imperfect private read on that trajectory: $c_{i,t+1} = \theta_{t+1} + \varepsilon_i$, with ε_i i.i.d. across agents and independent of θ_{t+1}.

Locking also imposes a private cost φ_i, i.i.d. across agents and independent of $(\theta_{t+1}, \varepsilon_i)$. For coins, φ_i is purely a liquidity cost, reflecting each agent's individual demand for transactional flexibility. For governance tokens, $\varphi_i = \ell + \kappa_i$ decomposes into a common lock-induced illiquidity cost $\ell \geq 0$—positive for vote-escrow protocols that require an explicit time lock (e.g., Curve's veCRV, Balancer's veBAL), zero for protocols whose governance requires no lock (e.g., Uniswap, Compound)—and an idiosyncratic engagement cost $\kappa_i \geq 0$, the time and attention cost of participating in governance, i.e., exercising *voice* rather than *exit* (Hirschman, 1970; Edmans and Manso, 2011). Agent i locks if and only if

$$b_t + \theta_{t+1} + \varepsilon_i \geq \varphi_i \quad \Longleftrightarrow \quad \eta_i \leq b_t + \theta_{t+1}, \qquad \eta_i \equiv \varphi_i - \varepsilon_i.$$

Because η_i is i.i.d. across the continuum of agents with fixed CDF H (the convolution of the cost distribution and the private-signal noise distribution), the strong law of large numbers implies that the realized cross-sectional locking ratio converges almost surely to

$$\lambda_t = H(b_t + \theta_{t+1}),$$

a deterministic, strictly increasing function of the asset's true future growth trajectory θ_{t+1}, given the publicly observed current benefit b_t. Idiosyncratic noise in individual costs and private signals washes out in the aggregate; what survives is exactly the cross-sectional average of private information about θ_{t+1}. This is the precise sense in which λ_t is a publicly verifiable signal of the holder base's aggregate conviction about the asset's future—consistent with the monetary search-theoretic view of λ as a coordination signal (Kiyotaki and Wright, 1993; Lagos and Wright, 2005) for coins, and with the costly-signaling logic of Spence (1973) for governance tokens—and it is a derived property of the model above, not an assumption asserted on λ_t directly. The coin/token distinction enters only through the composition of b_t (seigniorage versus fee-share-or-zero) and of φ_i (pure liquidity cost versus lock cost plus engagement cost); the threshold-crossing logic, and the resulting formula for λ_t, is identical across both.

That λ_t aggregates private information about θ_{t+1} does not imply that θ_{t+1} is already priced at t. Recovering θ_{t+1} from λ_t requires netting out the observed benefit b_t and inverting H, which in turn requires knowing the population distribution of locking costs and signal noise—itself not common knowledge, and, for governance tokens, requires that private beliefs be revealed only through the costly action of locking rather than through costless disclosure. Only a fraction of market participants perform this extraction, so, as in Grossman and Stiglitz (1980), costly information acquisition prevents λ_t from being fully impounded into price at t. As θ_{t+1} materializes and becomes public over $t+1$, price converges toward the fundamental value already anticipated by the high-λ holder base, so realized returns are higher for assets with higher λ_t. This partial-revelation step is the one substantive assumption in the model—common to both asset classes—and we test its return implication directly as Proposition 1.

Proposition 1. *$Cov_j(r_{j,t+1}, \lambda_{j,t}) > 0$, where j indexes assets and the covariance is taken in the cross-section at a given date: assets with a higher locking ratio, and hence higher SoV/MoE ratio $\lambda_t/(1-\lambda_t)$, earn higher subsequent returns, for both coins and governance tokens.*

This is Hypothesis 1, restated formally as an empirical proposition, separately for the two compositions of b_t and φ_i:

Hypothesis 1a (Coins). *For native coins with a seigniorage- or security-staking-eligible design, assets with higher $\lambda/(1-\lambda)$ in month t earn higher returns over months $t+1$ through $t+h$, after controlling for size, momentum, and market beta. This mechanism is architecture-agnostic: it applies to any native coin with a staking-for-security role, whether issued on an*

L1 or an L2.

Hypothesis 1b (Governance Tokens). *For DeFi governance tokens, assets with higher governance staking $\lambda/(1-\lambda)$ in month t earn higher returns over months $t+1$ through $t+h$, after controlling for protocol category, size, and momentum. Voting-weighted λ has incremental predictive power beyond passive staking λ alone.*

The seigniorage channel (Cong and He, 2024) reinforces Hypothesis 1a: because b_t enters the locking threshold directly for coins, staking is a revealed preference that seigniorage income plus anticipated price appreciation exceeds the opportunity cost of locking capital—a rational decision rule rather than a behavioral artifact. Whether a coin's native asset is issued on an L1 or an L2 is immaterial to this mechanism; what matters is only whether locking the asset earns $b_t > 0$. We classify individual assets into Hypothesis 1a (coin-like, $b_t > 0$ from a staking-for-security design) versus Hypothesis 1b (token-like, governance-only) by this functional criterion in Section 3, rather than by network layer.

The voting-weighted extension in H1b follows directly from the same model. Suppose active voting requires an additional monitoring cost $\psi_i > 0$ beyond the cost of locking alone (passive staking, or auto-compounding, requires no ongoing engagement). The threshold for voting, $\eta_i + \psi_i \leq b_t + \theta_{t+1}$, is strictly tighter than the threshold for locking alone, so voters are a more selected, higher-conviction subset of lockers. Voting-weighted λ therefore carries incremental information about the holder base's average conviction beyond passive locking λ alone—the empirical test of the Spence costly-signaling logic embedded in the second sentence of H1b.

The cost structure $\varphi_i = \ell + \kappa_i$ nests both institutional forms locking takes for governance tokens in practice. For vote-escrow protocols that require an explicit lock, $\ell > 0$ is a real, continuously variable illiquidity cost that scales with chosen lock duration, and locked supply is the natural empirical proxy for λ_t. For protocols whose governance is decided by simple token-balance voting with no lock, $\ell = 0$: liquidity is never forfeited, and κ_i, the cost of monitoring the protocol and casting an informed vote, is the only force sustaining the threshold-crossing equilibrium. The threshold logic is identical in both cases, but the observable action revealing $\eta_i \leq b_t + \theta_{t+1}$ differs: locked supply where a lock mechanism exists, and otherwise the joint behavior of holding without selling—forgoing the option to exit, in the spirit of Hirschman (1970)—and active voting participation. We therefore treat λ_t as an empirical index over whichever conviction-revealing channels are observable for a given asset—staking/locking, holding duration, and voting engagement—with the specific construction and channel weighting determined by data availability and detailed in Section 3.

2.2 Price Discovery and the Conditioning Role of Growth-Levelized NVT

Section 2.1 establishes that λ_t carries private information not yet fully reflected in price. The remaining question is what determines *how much* of that information remains unpriced at a given date. Let $\delta_t \in [0, 1]$ denote the fraction of the conviction signal embedded in λ_t that has already been absorbed into the current price p_t, so that $p_t = p_t^{\text{fund}}(1 + \delta_t)$ for some notion of fundamental value p_t^{fund} consistent with current economic activity. Holding fundamentals fixed, a higher current price relative to current transaction-based fundamental value corresponds to a higher δ_t; this is exactly what our Growth-Levelized NVT ratio measures, since $\text{NVT}_{GL,t} = \text{MC}_t/PQ_t^*$ is increasing in price for fixed PQ_t^*. The expected return earned from t to $t + 1$ as the conviction signal is gradually validated and absorbed into price is proportional to the *unabsorbed* portion of the signal, $\lambda_t(1 - \delta_t)$, rather than to λ_t alone. It follows that the marginal predictive power of λ_t on returns is decreasing in δ_t, and therefore decreasing in $\text{NVT}_{GL,t}$.

Proposition 2. *The cross-partial effect of λ_t and $NVT_{GL,t}$ on expected returns is negative: the return predictability of $\lambda_t/(1 - \lambda_t)$ is concentrated among assets with low $NVT_{GL,t}$.*

Hypothesis 2. *The return predictability of $\lambda/(1 - \lambda)$ documented in H1 is significantly stronger among assets with below-median Growth-Levelized NVT ratios. Equivalently, the marginal return to a unit increase in $\lambda/(1 - \lambda)$ is higher when the asset is cheap relative to* *its fundamental economic activity.*

This is the central insight motivating the two-dimensional quadrant framework: $\lambda/(1-\lambda)$ supplies the conviction signal, while Growth-Levelized NVT indicates how much of that signal the market has already absorbed. Unlike $\lambda/(1-\lambda)$, Growth-Levelized NVT carries no independent theoretical prediction for returns in our framework—a low Growth-Levelized NVT ratio alone, without a high conviction signal, simply describes an asset with limited current economic activity, not necessarily one that is mispriced. Empirically, H2 implies a statistically significant interaction between $\lambda/(1-\lambda)$ and Growth-Levelized NVT in cross-sectional return regressions, with the coefficient on $\lambda/(1-\lambda)$ larger in absolute value for below-median Growth-Levelized NVT assets.

2.3 The Quadrant Portfolio

Propositions 1 and 2 jointly imply that the assets with the highest expected unrealized returns are those combining a strong, on-chain-verified conviction signal with valuation that has not yet absorbed it: high $\lambda/(1-\lambda)$ and low Growth-Levelized NVT. We label this combination *Star*. The mirror-image combination—low $\lambda/(1-\lambda)$ and high Growth-Levelized NVT, signaling both weak conviction and a valuation already stretched beyond current fundamentals—we label *Avoid*.

Hypothesis 3. *A long-short portfolio that goes long Star assets (above-median $\lambda/(1-\lambda)$, below-median Growth-Levelized NVT) and short Avoid assets (below-median $\lambda/(1-\lambda)$, above-median Growth-Levelized NVT), rebalanced monthly, generates positive and statistically significant alpha after controlling for market, size, and momentum factors. The Stars-minus-Avoid Sharpe ratio significantly exceeds that of equal-weighted benchmark portfolios.*

H3 is the integrating test of the framework: it asks whether the joint signal has incremental predictive power beyond either dimension of Propositions 1–2 alone. The alpha estimate quantifies the economic magnitude of the mispricing the quadrant framework identifies, net of compensation for systematic risk exposures. We report portfolio results separately for coins and governance tokens to preserve the distinction between the seigniorage-driven locking mechanism for coins and the engagement-cost-driven locking mechanism for governance tokens, both nested within Proposition 1, and test whether the alpha is larger for governance tokens, where the illiquidity and engagement costs $\ell + \kappa_i$ create a higher signaling barrier than the seigniorage-driven locking decision for coins.

3 Data and Variable Construction

[To be written.]

4 Results

[To be written.]

5 Robustness

[To be written.]

6 Conclusion

[To be written.]

A Derivation of the SoV/MoE Decomposition

This appendix provides the formal derivation of the SoV/MoE decomposition and its simplification to a function of λ alone.

Setup. Let M denote the total circulating supply of a digital asset and MC its market capitalization, which serves as the money supply proxy in the QTM framework. Let PQ

denote nominal economic output (on-chain transaction volume for coins; protocol throughput for tokens). Standard QTM gives $\mathrm{MC} \cdot V = PQ$, so the market capitalization satisfies $\mathrm{MC} = PQ/V$.

The locking ratio. Define λ as the proportion of total supply that is locked or staked:

$$\lambda = \frac{M_{\text{locked}}}{M}, \qquad \lambda \in [0, 1).$$

The free-circulating supply is $M(1-\lambda)$. We define the *free velocity* V_{free} as the velocity of only the economically active (unlocked) portion:

$$V_{\text{free}} = \frac{PQ}{M(1-\lambda)}.$$

MoE and SoV components. The Medium of Exchange component of market capitalization is the value attributable to current transactional activity, priced at the free velocity:

$$\mathrm{MC}_{\text{MoE}} = \frac{PQ}{V_{\text{free}}} = M(1-\lambda).$$

The Store of Value component is the residual:

$$\mathrm{MC}_{\text{SoV}} = \mathrm{MC} - \mathrm{MC}_{\text{MoE}} = M - M(1-\lambda) = M\lambda.$$

The SoV/MoE ratio. The ratio of the SoV to MoE components is:

$$\frac{\mathrm{MC}_{\text{SoV}}}{\mathrm{MC}_{\text{MoE}}} = \frac{M\lambda}{M(1-\lambda)} = \frac{\lambda}{1-\lambda}.$$

This result shows that the SoV/MoE ratio is a monotone increasing function of λ alone, requiring no estimation of velocity, transaction volume, or any other derived quantity. As $\lambda \to 0$, the ratio approaches zero (pure medium of exchange). As $\lambda \to 1$, the ratio diverges (pure store of value).

The Growth-Levelized NVT ratio. The Network Value to Transactions ratio is NVT = MC$/PQ_{\text{annualized}}$. We construct a forward-looking version by replacing current PQ with PQ^*, the levelized present value of the projected transaction stream:

$$PQ^* = \frac{\sum_{s=1}^{n} \frac{PQ_0(1+g)^s}{(1+r_e)^s} + \frac{PQ_0(1+g)^n(1+g_\infty)}{(r_e-g_\infty)(1+r_e)^n}}{\text{annuity factor}(r_e, n)},$$

where g is the trailing k-year CAGR of PQ, r_e is the CAPM-estimated required return, g_∞ is the terminal growth rate, and the annuity factor converts the present value to a levelized annual equivalent (analogous to the Excel PMT function). The Growth-Levelized NVT ratio is then $\text{NVT}_{GL} = \text{MC}/PQ^*$. Full estimation details are in Section 3.

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